

RAG Accuracy Report — Week 4

Zynxis Agentic AI Internship — RAG for Agents Author: Faozan Mujtaba

1. Setup

Knowledge base: 3 PDFs from earlier weeks' own deliverables (Week 1's `logic_flow_diagram.pdf`, Week 3's `tech_note.pdf` and `architecture_diagram.pdf`) — 5 pages total. Reusing known documents means every ground-truth fact used below is independently verifiable against the source PDF.

Ingestion (`ingest.py`): each PDF page is extracted with `pypdf`, then split into ~120-word chunks with a 20-word overlap. 12 chunks were produced from the 5 source pages.

Retrieval (`rag.py`): ChromaDB's default embedding function, top-k=4 nearest chunks by similarity, injected into the prompt as labeled `[source p.N]` context blocks. The model (Groq `llama-3.3-70b-versatile`) is instructed to answer only from that context and to cite its source.

2. Methodology

Each of 10 test questions is scored on two independent axes:

- **Retrieval accuracy** — did the expected source PDF appear anywhere in the top-4 retrieved chunks?
- **Generation accuracy** — does the final answer contain every expected keyword (case-insensitive substring match)?

This separates "did we fetch the right passage" from "did the model use it correctly," which a single pass/fail score would hide.

3. Results

#	Retrieval	Generation	Question
1	PASS	PASS	Long-term memory vector DB + persist path
2	PASS	PASS	Why ChromaDB over Pinecone
3	PASS	PASS	Short-term buffer size (N turns)
4	PASS	PASS	Suggested fix for stale-fact relevance
5	PASS	PASS	LLM model in the architecture diagram
6	PASS	PASS	Zebra puzzle: animal Bob owns
7	PASS	PASS	Zebra puzzle: Alice's house color
8	PASS	PASS	CoT's typical failure mode
9	PASS	PASS	Does CoT use external tools
10	PASS	PASS	What "ReAct = ?" per the Key Insight box

Retrieval accuracy: 10/10 (100%) Generation accuracy: 10/10 (100%)

Full transcripts (question, retrieved chunks, generated answer) are in `eval_results.json`, produced by running `python eval.py --json eval_results.json` against the live pipeline.

4. Observations & limitations

- A perfect score reflects the benchmark's scale, not a claim that this pipeline is error-free at production scale: n=10 questions over a 5-page, 12-chunk corpus is small enough that top-4 retrieval essentially always surfaces the right page.
- Keyword substring matching is a strict, cheap, and reproducible proxy for correctness — it doesn't need an LLM judge — but it doesn't verify reasoning quality on more open-ended questions. Longer answers with paraphrased facts (e.g. "the failure mode is generating incorrect text mid-reasoning" instead of the literal word "hallucination") would be marked wrong even if substantively correct.
- The real stress test for this pipeline would be a much larger, less redundant corpus (dozens of pages covering overlapping topics), where chunk boundaries and top-k choice start to matter for retrieval accuracy. That's a natural next step beyond this week's scope.